

TECHNOLOGY NEWS



Python 3.0 is now intentionally backwards incompatible

Python developers have released the final version of Python 3.0 (also called Python 3000 or Py3k), a major reworking of the programming language that is incompatible with the Python 2 series. The language is mostly the same, but many details, especially how built-in objects like dictionaries and strings work, have changed considerably, and a lot of deprecated features have finally been removed.



Also, the standard library has been reorganized in a few prominent places, developers said.

In a document outlining the changes, Guido van Rossum, creator, Python, said, "Nevertheless,

after digesting the changes, you'll find that Python really hasn't changed all that much—by and large, we're mostly fixing well-known annoyances and warts, and removing a lot of old cruft."

The `print` statement has been replaced with a `print()` function, with keyword arguments to replace most of the special syntax of the old `print` statement (PEP 3105).

Another major change is that Unicode will now be the default. Python 3.0 uses the concepts of text and (binary) data instead of Unicode strings and 8-bit strings. All text is Unicode; however encoded Unicode is represented as binary data. The type used to hold text is `str`, the type used to hold data is bytes. The biggest difference with the 2.x situation is that any attempt to mix text and data in Python 3.0 raises `TypeError`, whereas if you were to mix Unicode and 8-bit strings in Python 2.x, it would work if the 8-bit string happened to contain only 7-bit (ASCII) bytes, but you would get `UnicodeDecodeError` if it contained non-ASCII values.

Hackable:1, a new distro for hackable devices

A new distribution for the Neo and other hackable devices dubbed Hackable:1 has been released. Based on the DebianOnFreerunner, it packages the OM2007.2 applications, extending and bug-fixing them. It is intended to become a stable platform for the VAR market and is fun to use for everybody else. Some of the highlights include: OM2007.2 packaged as `.deb` packages that include the dialler, SMS, contacts, neod, phone-kit, gsmcd, matchbox and panel applets; improved sound quality (fixes for gsmcd for echo cancellation); extended AUX and power menus; simple onscreen keyboard with all hacker characters on a short press on the AUX button; GPS works out of the box; many GPRS providers preconfigured for easy use; and GSM multiplexing preconfigured (that is, you can have calls and SMSs coming in during a GPRS session).

The distro comes as a tarball and you can download it from www.hackable1.org/hackable1/?C=M;O=D. In order to get started, you'll require a 2 GB SD card and a card reader for your PC/laptop. Partition and format the SD card, and then simply untar the tarball onto it. Your Flash even remains untouched, so you can easily give it a test run. For more information, check out www.hackable1.org.



Movial Octopus: A central point of contact for all multimedia requirements

Movial has announced it is contributing the Movial Octopus Media Engine, the multimedia enabling source code, to the mobile Linux community. Octopus uses the OpenMAX standard and enables easy integration of multimedia into different mobile applications.

The Movial Octopus Media Engine controls audio and video content that can be read from local files or streamed over the network. Octopus provides a higher-level API for end-user applications to manage multimedia content. Target applications include media players as well as voice and video call applications for devices such as MIDIs and Netbooks. Octopus works as a background service that several applications can use simultaneously. For media content operations, such as video calls, Internet streaming and MP3 playback, Octopus uses either GStreamer or OpenMAX IL components. Developers can download Octopus at sandbox.movial.com/wiki/index.php/Octopus. The current client API is a D-Bus API and plans are underway to offer an OpenMAX AL API in 2009.

Novell's new PlateSpin supports leading hypervisors

Novell has enhanced its PlateSpin Workload Management solution. The new PlateSpin Recon, PlateSpin Migrate, PlateSpin Protect and PlateSpin Orchestrate enable users to profile, migrate, protect and manage server workloads between physical and virtual infrastructures in heterogeneous IT environments. PlateSpin Workload Management, according to the company, is the only solution on the market today to support 32- and 64-bit Linux and Windows servers, as well as all leading hypervisors.